

KAI 9

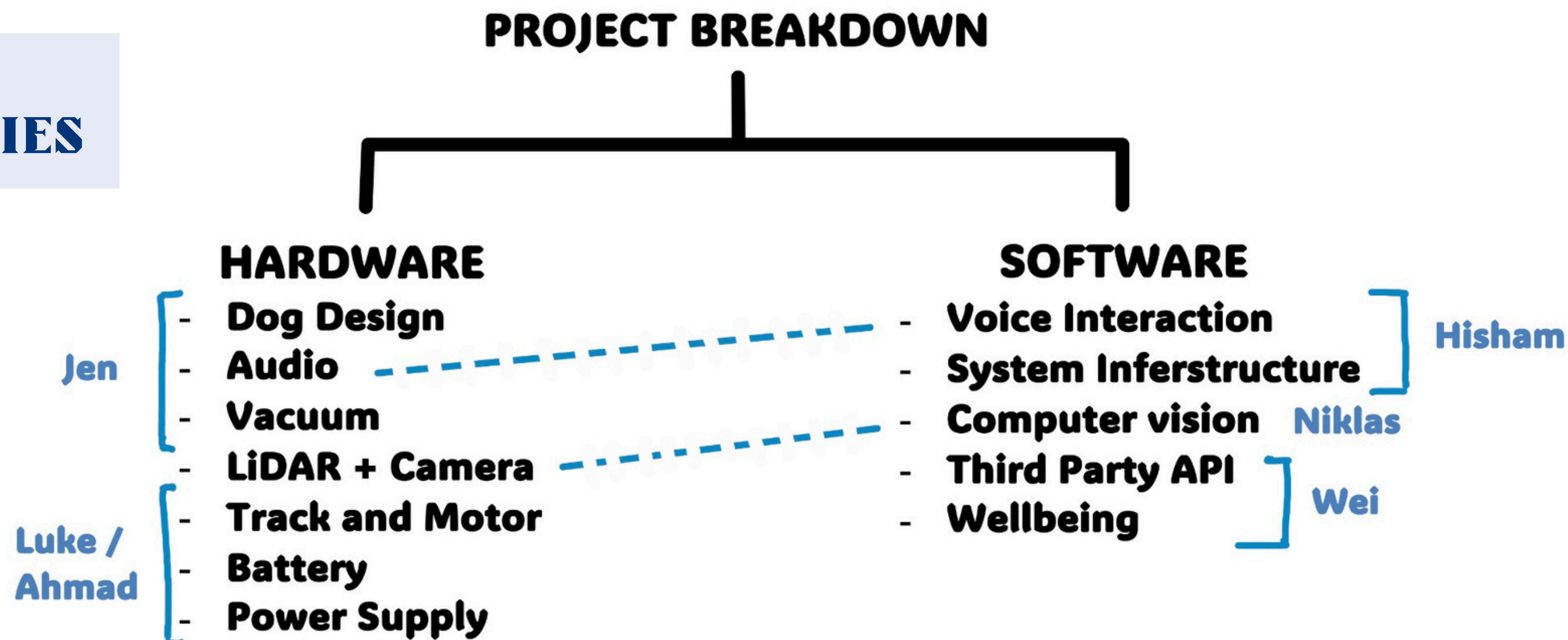


**MID POINT
PROJECT UPDATE**

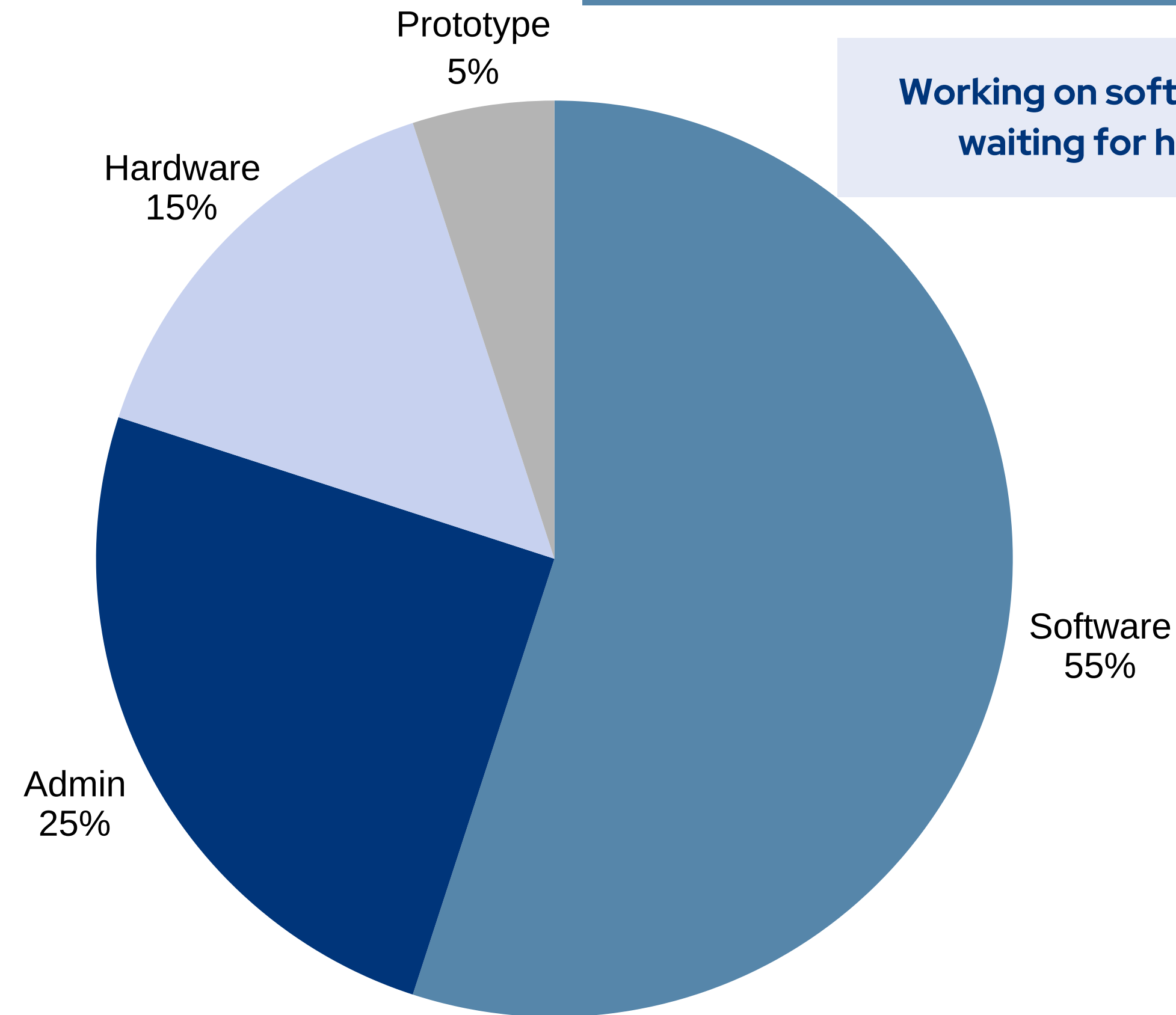
PROJECT TIMELINE



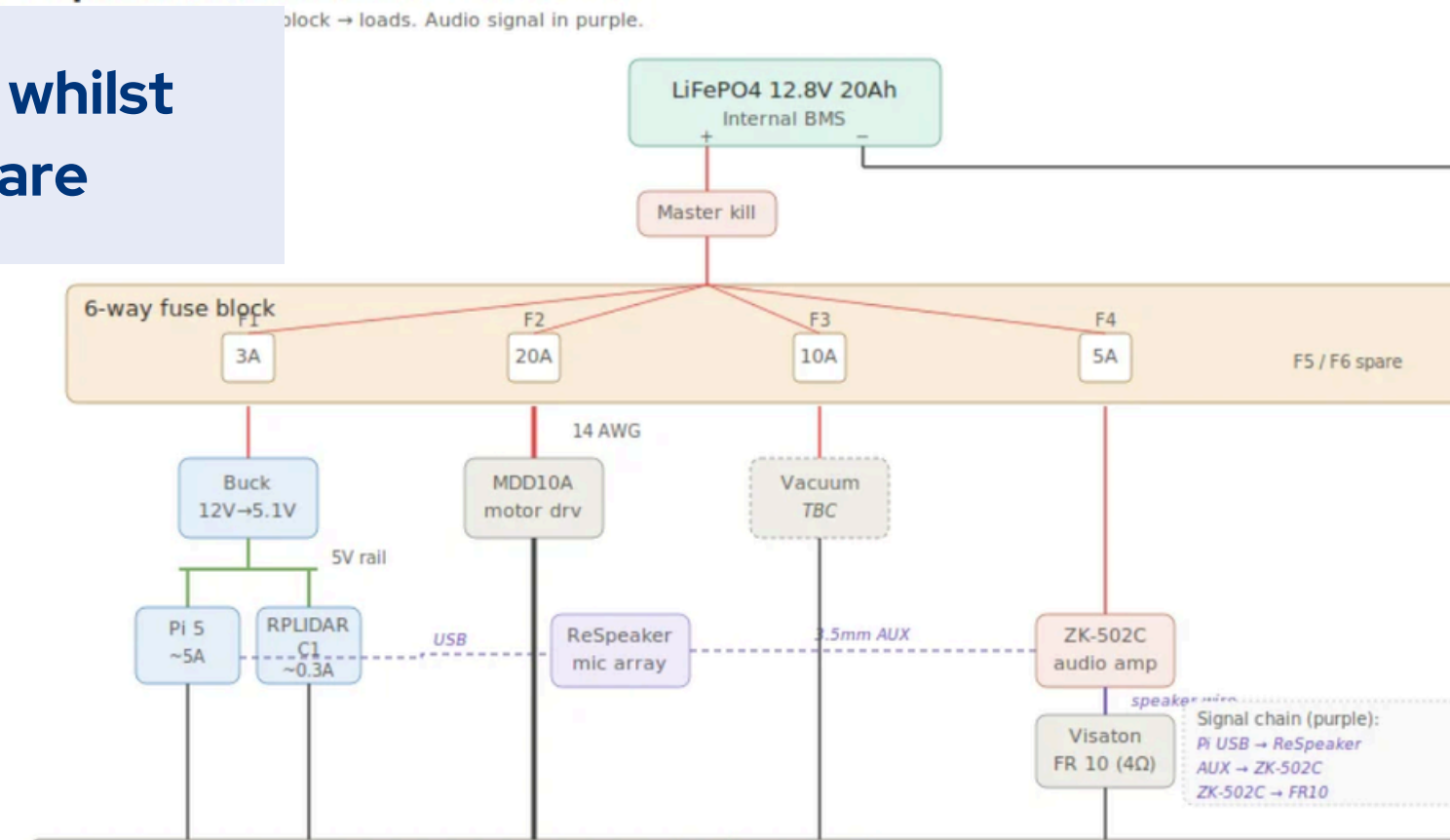
TEAM RESPONSIBILITIES



WHAT HAVE WE DONE ?



Full power distribution & audio chain



			Total	£	658.02
			Budget	£	1,300.00
			Money Left	£	641.98
ARRIVED	Component	Quantity	Price (£)	Link	
	LiDAR	1	£	https://uk.robotshop.com/products/slamtec-rplidar-c1	
Y	Camera	1	£ 65.00	https://uk.rs-online.com/web/p/raspberry-pi-cameras/	
	Battery	1	£ 65.00	https://azeco.co.uk/product/lithium-battery-lifepo4-20	
Y	Mic	1	£ 50.55	https://uk.farnell.com/seeed-studio/107990193/usb-m	
	Speaker	2	£ 12.95	https://uk.farnell.com/visaton/fr10-2020/speaker-full-r	
	Vacuum	1	£ 85.00	https://www.argos.co.uk/product/5797318	
Y	Dc motor driver	1	£ 24.90	https://thepihut.com/products/10a-5v-30v-2-channel-d	
Y	Motors	2	£ 15.90	https://thepihut.com/products/metal-dc-geared-motor	
Y	Brackets for motors	1	£ 11.50	https://thepihut.com/products/pololu-stamped-alumin	
Y	Mounts for motors	1	£ 12.50	https://thepihut.com/products/pololu-universal-alumir	
	battery charger	1	£ 64.99	https://uk.eco-worthy.com/products/5a-10a-12v-smart	
Y	Audio Amplifier	1	£ 11.65	https://uk.farnell.com/dfrobot/dfrobot0803/bluetooth-pow	
	LiDAR Adapter	1	£	https://thepihut.com/products/cp2102-usb-uart-board	

SETTING UP BOB ACCOUNTS



ADMIN UPDATE

ENTRY 01 • SOFTWARE TEAM

BOB BLOG

Getting the Robot to Listen

Before a robot can help someone, it needs to hear them. That sounds obvious, but getting reliable speech recognition working — in real time, on low-power hardware, across different accents and languages — turned out to be the first real engineering challenge of this project. This entry covers how we approached it.

LEAFLET

”

Mum still won't use a smartphone. But she talks to Kai every morning — and he reminds her to take her walk.

— ANNA R., DAUGHTER & CARER

1 in 3

people over 65 in the UK live alone. Half say they go a week without a meaningful conversation.

—
Old age shouldn't mean losing your independence.

Kai 9 is a small, friendly robot designed to live alongside an older adult — keeping their floors clean, their phone within reach, and their day quietly in motion. Not a gadget. A companion.

KINDRED AI • MEET

Kai9

A loyal little friend who *helps around the house.*

BADGES



Enterprise Design Thinking Practitioner

ISSUED TO
Ahmad Wahab



Issued on: 08 MAY 2025 | Issued by:
Verify: <https://www.credly.com/go/>

Classifying Data Using
IBM Granite

IBM SkillsBuild

Turn Ideas into Reality
with Vibe Coding



IBM SkillsBuild

API INTERGRTION

Command → Parser → Safety Check → Service Module → Robot Response

```
def handle_message(user_text):
    text = user_text.lower().strip()

    if text in ["weather", "today's weather"]:
        return weather_service.get_update()

    elif text in ["play music", "play relaxing music"]:
        return spotify_service.play_playlist("relaxing")

    elif text in ["activity", "activity today"]:
        return activity_tracker.get_daily_summary()

    elif text in ["stop", "emergency stop", "halt"]:
        robot.safety_stop()
        return "Emergency stop activated."

    else:
        return "Sorry, I did not understand. Try: weather, pl
```

```
AUTHORIZED_USERS = ["+447123456789"]

MAX_MOVE_TIME = 3.0
LOW_BATTERY_LIMIT = 20

def safety_check(command, sender, battery_level):
    if command == "stop":
        return True, "Emergency stop allowed."

    if sender not in AUTHORIZED_USERS:
        return False, "Command rejected: user not authorised."

    if battery_level < LOW_BATTERY_LIMIT:
        return False, "Command rejected: battery too low."

    if command.startswith("walk"):
        duration = float(command.split()[-1])

        if duration > MAX_MOVE_TIME:
            return False, "Command rejected: movement time too long."

    return True, "Command approved."
```

speech_to_text.py 4 X

C: > Users > hisha > OneDrive > Desktop > Kai 9 EOY > speech_to_text.py > ...
22 parser.add_argument("--list-devices", action="store_true", help="List audio devices and exit")
23 parser.add_argument("--device", type=int, default=None, help="Audio device index")
24 parser.add_argument("--model", default="base",
25 | | | | | choices=["tiny", "base", "small", "medium", "large-v3"],
26 | | | | | help="Whisper model size (default: base)")
27 parser.add_argument("--output", default=None, help="File to append final transcriptions to")
28 parser.add_argument("--language", default=None, help="Force language code e.g. 'en', 'ar'")
29 args = parser.parse_args()
30
31 # List devices (no RealtimeSTT needed for this)
32
33 if args.list_devices:
34 | import sounddevice as sd
35 | devices = sd.query_devices()
36 | print("\nAvailable audio INPUT devices:")
37 | print(f" {'#':<4} {'Max ch':<8} Name")
38 | print(" " + "-" * 50)
39 | for i, d in enumerate(devices):
40 | | if d["max_input_channels"] > 0:
41 | | | print(f" {i:<4} {int(d['max_input_channels']):<8} {d['name']}")
42 | print()
43 | sys.exit(0)
44
45 # Find ReSpeaker if no device specified
46
47 if args.device is None:
48 | import sounddevice as sd
49 | for i, d in enumerate(sd.query_devices()):
50 | | if "respeaker" in d["name"].lower() or "seeed" in d["name"].lower():
51 | | | if d["max_input_channels"] > 0:
52 | | | | args.device = i
53 | | | | print(f"✓ Auto-detected ReSpeaker: [{i}] {d['name']}")
54 | | | | break
55 | if args.device is None:
56 | | print("⚠ ReSpeaker not found ⚠ using system default input.")

SPEECH RECOGNITION

Using Whisperflow library,
integrating live transcription
for faster response.

C:\Users\hisha\OneDrive\Desktop\Kai 9 EOY>python speech_to_text.py --device 1
Loading Whisper 'base' model ... (first run downloads it)
✓ Model ready.

Using device [1]

🎤 Listening ... speak naturally. Ctrl+C to stop.

[10:38:45] Hello, this is Hisham texting the transcribing abilities of my microphone.
[10:39:06] Yeah, it got texting instead of testing But this time it got testing right.
RealtimeSTT shutting down

— Session ended —

LIDAR + CAMERA

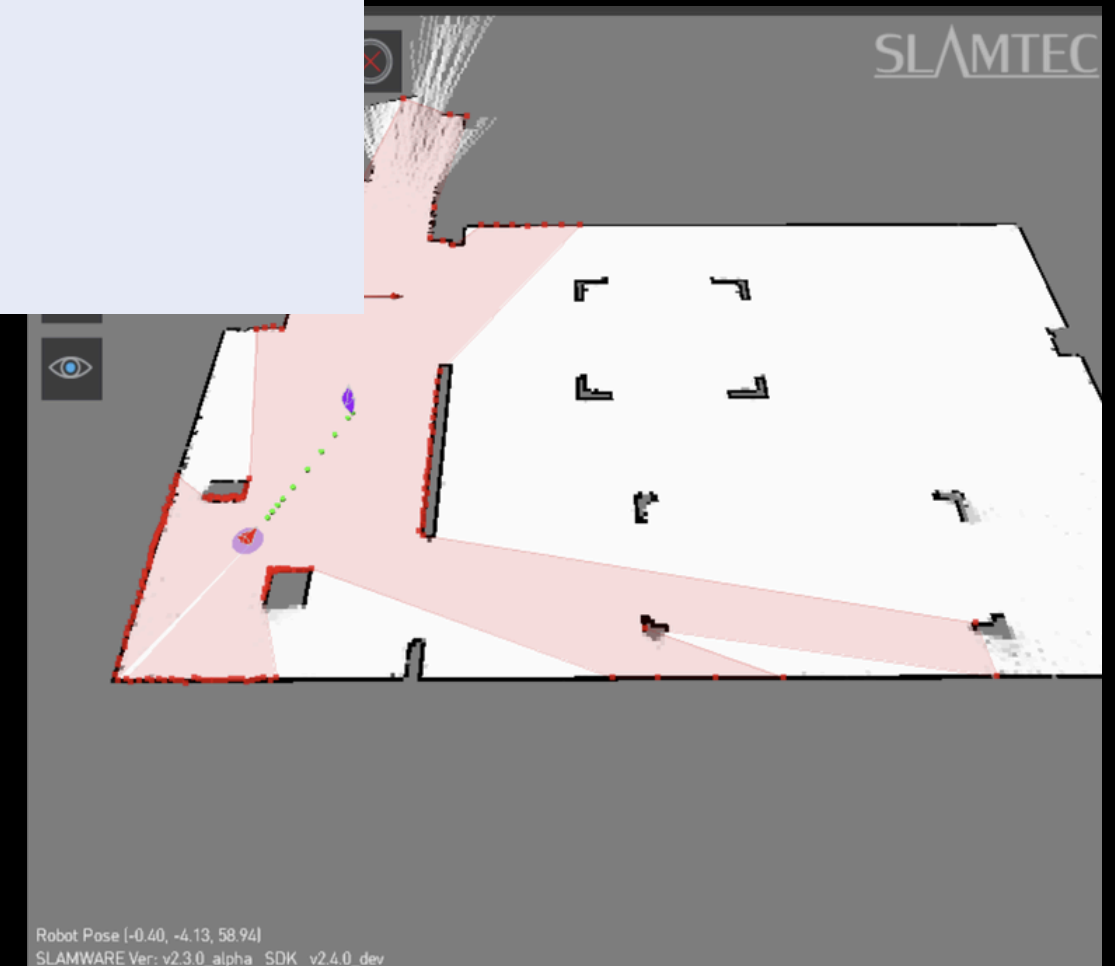
SLAM

```
lidar: RPLidar = RPLidar(port=PORT_NAME, baudrate=115200)

def process_scan(scan) -> dict[str, float]:
    zones: dict[str, float] = {
        'front': float('inf'),
        'left': float('inf'),
        'right': float('inf'),
        'back': float('inf')
    }

    for (quality, angle, distance) in scan:
        if distance == 0:
            continue

        if angle >= 315 or angle < 45:
            zones['front'] = min(zones['front'], distance)
        elif 45 <= angle < 135:
```

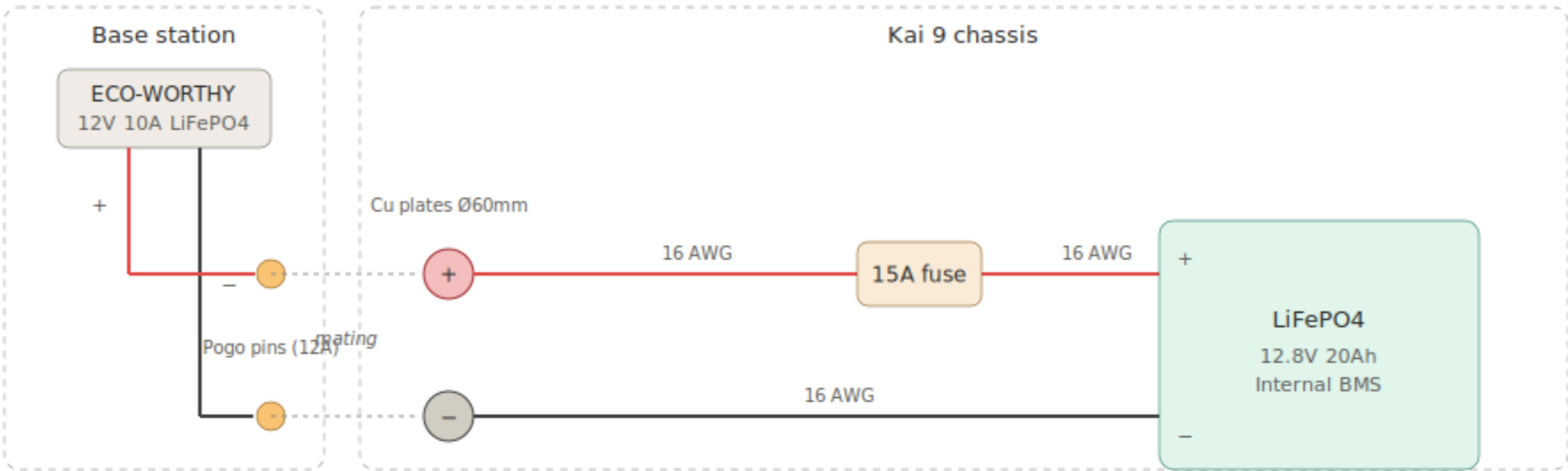


OBJECT DETECTION FOR OBSTACLE AVOIDANCE



Phase 1: Direct-wire charging

Bench-test first. Relay isolation deferred to Phase 2.



Bench-test checklist

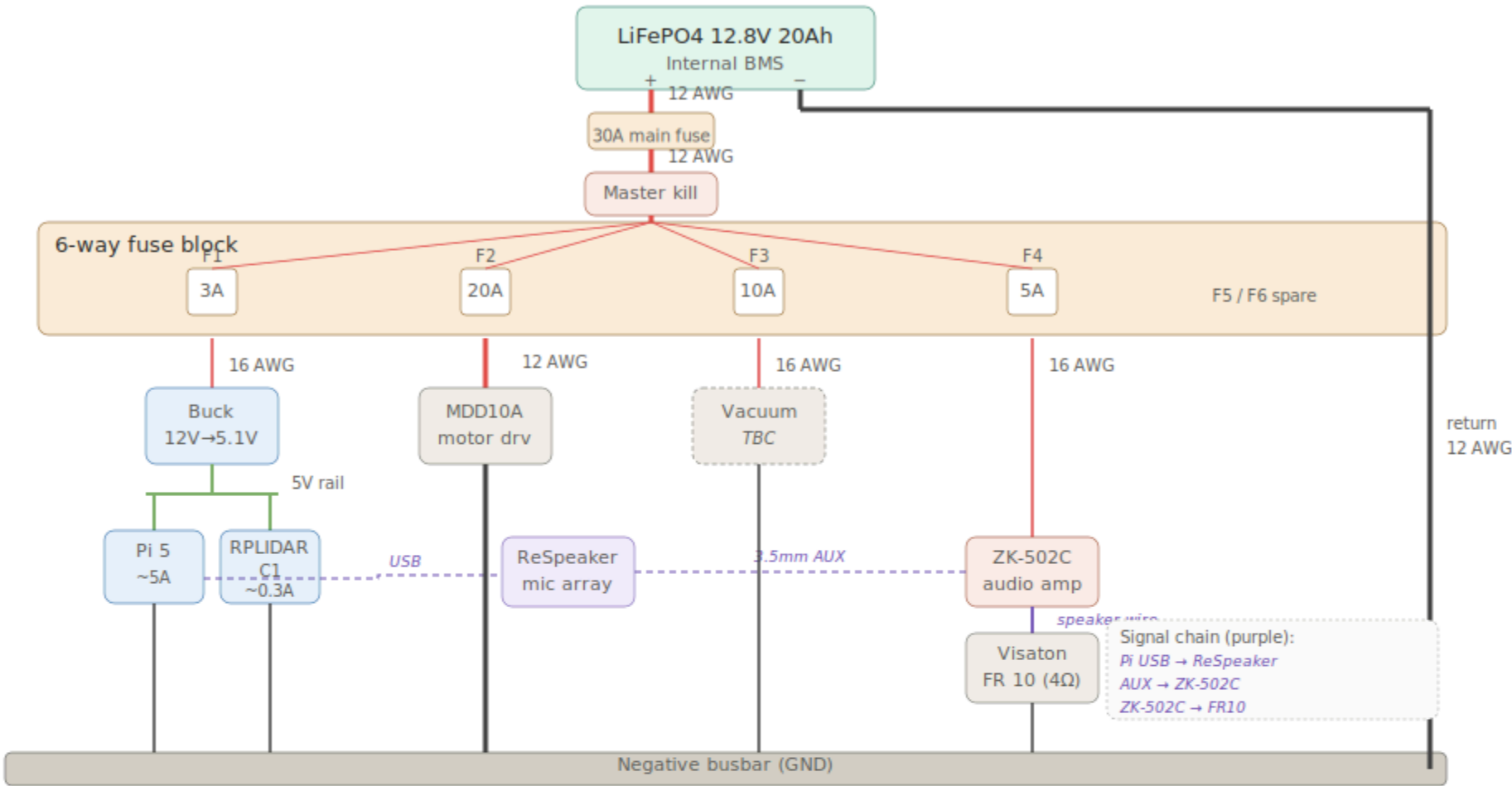
- Polarity verified with multimeter every cycle
- 15A fuse blow test confirmed before chassis install
- 20+ docking cycles — check for plate pitting
- Charge profile logged (target ~3 h to full)
- Plate-to-fuse junction uses spade connectors
- Leave 50×50 mm flat space for Phase 2 relay

Bill of materials & fuse map

Component	Part / model	Fuse	Wire	~Cost
Battery	AZECO 12.8V 20Ah LiFePO4 (BMS)	—	12 AWG	£64.99
Charger	ECO-WORTHY 12V 10A smart LiFePO4	15A (chg)	16 AWG	£64.99
Dock contacts	Mill-Max pogo 0868 + Cu plates Ø60	—	—	£9.10
Main fuse (load)	30A ATO blade + RS PRO inline holder	30A	12 AWG	~£5
Master kill	Blue Sea / Carling 30A DC toggle	—	12 AWG	~£15
Fuse block	6-way ATO/ATC + busbar (BlueSea 5025)	—	—	£30
F1: Buck→Pi+LiDAR	XL4016E1 (5.1V, set via pot + meter)	3A	16 AWG	~£6
F2: Motor driver	Cytron MDD10A (2×10A cont, 30A pk)	20A	12 AWG	£25
F3: Vacuum	TBC — 12V car handheld (likely)	10A*	16 AWG*	~£20*
F4: Audio amp	ZK-502C HiFi Bluetooth amp (5-27V)	5A	16 AWG	£11.65
Mic array	Seeed ReSpeaker USB (via Pi USB)	—	USB cable	£50.55
Speaker	Visaton FR 10, 4Ω, 30 W continuous	—	speaker wire	£12.95

Full power distribution & audio chain

Battery → master kill → fuse block → loads. Audio signal in purple.



Key decisions & rationale

Phase 1 first: direct wire, no relay
Prove dock geometry, pogo pins, plate alignment, wire sizing, fuse blow behaviour before adding logic-controlled isolation. Slack wires + spade terminals at plate-to-fuse junction so Phase 2 relay drops in cleanly.

30A main fuse + Phase 2 relay = defence-in-depth
30A ATO main fuse within 150mm of battery+ protects upstream wire / master kill / fuse block input. Phase 2 adds SPST 40A relay between plate+ and battery+. Four protection layers: BMS, 30A main, relay isolation, branch fuses.

F2 motor branch: 20A fuse on 12 AWG
MDD10A is 10A continuous per channel, 30A peak. 15A nuisance-blows on inrush. 20A handles startup + stall bumps.

LiDAR & ReSpeaker ride the Pi 5V rail
LiDAR ~0.3A at 5V — sharing buck saves a converter, simplifies grounds. ReSpeaker is a USB peripheral of the Pi.

Vacuum: 12V car handheld preferred — choice TBC
Avoids 12V→25V boost (~15% runtime hit). MOSFET-switched from Pi GPIO. *F3 fuse/wire pending final part.

WHAT IS NEXT ?

SOFTWARE

- Move from individual feature demos to one connected, testable robot system.
- Create the digital architecture for the whole system and integrate it based off of our chosen compute device

WHAT IS NEXT ?

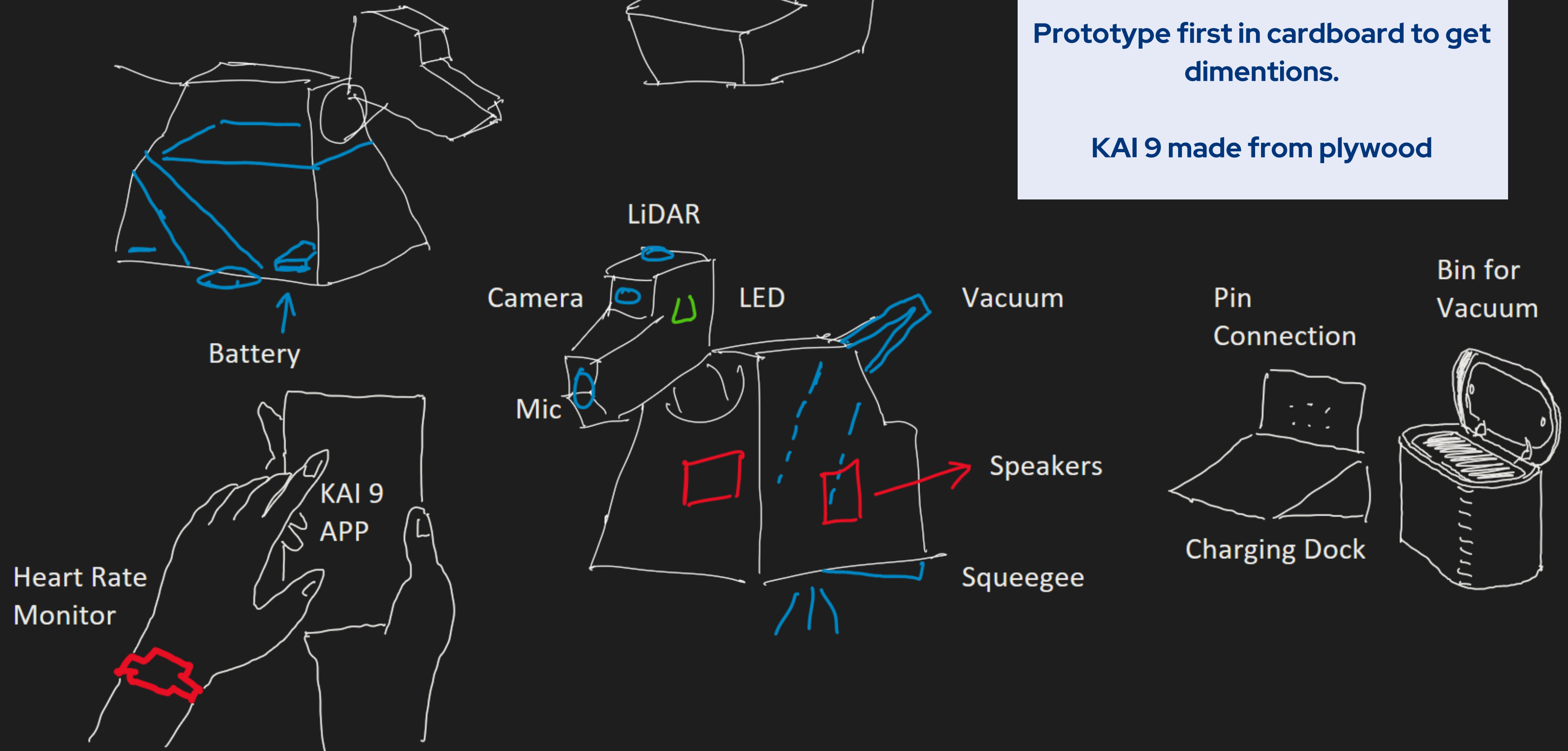
HARDWARE

- Assemble the main wooden chassis and wire the the motors to the Raspberry Pi
- Build the power managment and distribution system

PROTOTYPE

Prototype first in cardboard to get dimensions.

KAI 9 made from plywood

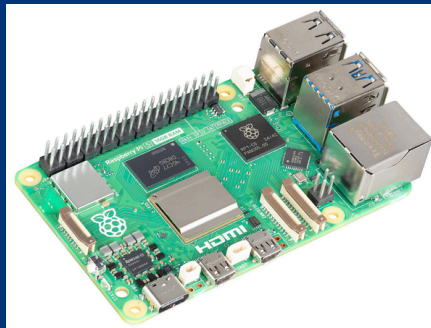


QUESTIONS ?



MAC MINI?

- Run everything locally including LLMs
- Upgraded server and network abilities
- Drawback: Power consumption, but can be optimized



MORE POWERFUL PI ?

- Room mapping via Lidar requires more memory
- Local data processing for privacy
- Suggestion: upgrade to raspberry pi 5, 4-16GB